Spec → Code: RID164

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1. Spec

REPORT ID

RID164

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

This report identifies Procedure overlaps based on matching PT/LLT terms and overlapping onset/end dates. (Report Title: Procedures Overlapping with same subject for same PT/LLTs)

COLUMN HEADING

1. Country
2. Site
3. Subject Number
4. Form
5. Encoding Hierarchy
6. PROC Encoded Term
7. PROC 1 Seq (PROC 1 is the one with smaller Seq. No. between the pair of overlapping records)
8. PROC 1 Verbatim
9. PROC 1 Start Date
10. PROC 1 Stop Date
11. PROC 2 Seq (PROC 2 is the one with bigger Seq. No. between the pair of overlapping records)
12. PROC 2 Verbatim
13. PROC 2 Start Date
14. PROC 2 Stop Date

LOGIC

Output is produced if a subject has overlapping Procedure Start Date and Procedure Stop Date for identical Procedure PT or LLT Terms. Each pair of overlapping Procedures should be output on one line.

User should select PT or LLT for comparison in parameter. If nothing is selected, output for overlapping identical LLT or PT.

Partial date also need to be considered while programming for identifying overlapping records

If either of the PROC has blank onset date or end date, consider them as ongoing.

If Procedure Start Date has UNK, please convert to the first day of the month and/or Jan;

If Procedure Stop Date has UNK, please convert to the last day of the month and /or Dec.

Deleted record and erroneous visits should not be considered.

For APPRC consider taking start date as stop date, since stop date is not available on form

REPORT SCENARIOS

Please check two forms: PROC and APPRC. Check within form only.

a. If parameter LLT is selected:

1) Output = YES
two identical Procedure terms (with same LLT after encoding) have overlapping start dates/end dates

2) Output = NO

identical Procedure terms (with same LLT after encoding) have no overlapping start dates/end dates

3) Output = NO
different Procedure terms (with different LLT after encoding)

4) Output = NO
Deleted PROC record

b. If parameter PT is selected, or no selection with PT/LLT or both selected:

1) Output = YES
two identical Procedure terms (with same PT and same LLT after encoding) have overlapping start dates/end dates

2) Output = YES
two identical Procedure terms (with same PT and different LLT after encoding) have overlapping start dates/end dates

3) Output = NO
identical Procedure terms (with same PT after encoding) have no overlapping start dates/end dates

4) Output = NO
different Procedure terms (with different PT after encoding)

5) Output = NO
Deleted PROC record

SORTING

Country, Site, Subject Number, PROC 1 Seq

2. Logic Understanding

This spec identifies pairs of Procedure records (within the PROC form, and separately within the APPRC form) for the same subject where the encoded term (LLT or PT, per user parameter selection) matches and the Start/Stop date ranges overlap, after applying date-cleaning rules for UNK values, partial dates, ongoing (blank date) records, and the APPRC-specific substitution of Start Date for the missing Stop Date. Matching criteria differ depending on whether LLT or PT (or default/both) is selected as the comparison parameter, with PT-level matching being more permissive (allowing different LLTs) than LLT-level matching. Each qualifying overlapping pair is output as one row, with the earlier-Seq record labeled PROC 1 and the later-Seq record PROC 2, sorted by Country, Site, Subject Number, and PROC 1 Seq, while deleted records and erroneous visits are excluded upfront.

1. Identify all Procedure records for a subject from the PROC form (and separately, from the APPRC form) - these two forms are checked independently of each other, per 'Please check two forms: PROC and APPRC. Check within form only.'

2. Exclude any deleted records and records tied to erroneous visits from consideration entirely, per 'Deleted record and erroneous visits should not be considered' and the scenario rule that Deleted PROC record always yields Output = NO.

3. For each subject, determine the comparison level (LLT or PT) based on the user-selected parameter: if LLT is selected, compare using the encoded LLT term; if PT is selected, or if no selection is made, or if both PT and LLT are selected, compare using the encoded PT term (per Report Scenarios section b heading covering 'no selection with PT/LLT or both selected'). *(inferred)*

4. Apply date-cleaning rules before comparing ranges: if Start Date contains UNK, convert the unknown day to the first day of the month and/or unknown month to January; if Stop Date contains UNK, convert unknown day to the last day of the month and/or unknown month to December.

5. If either record in a potential pair has a blank onset (start) date or blank end (stop) date, treat that record as ongoing (open-ended) for the purpose of the overlap check.

6. For APPRC records specifically, since there is no Stop Date field on that form, use the Start Date value as a substitute for the Stop Date when computing date ranges.

7. Account for partial (incomplete) dates generally when determining whether two date ranges overlap, per 'Partial date also need to be considered while programming for identifying overlapping records.'

8. Compare pairs of Procedure records within the same subject (and within the same form) to see whether their Start Date-to-Stop Date ranges overlap (intersect) in time.

9. If parameter LLT is selected: output YES only when two Procedure records share the same encoded LLT term AND their date ranges overlap; if the LLT terms differ, or if LLT terms match but dates do not overlap, output NO (per scenario a.1-a.3).

10. If parameter PT is selected (or no selection, or both PT/LLT selected): output YES when two records share the same encoded PT term and their dates overlap, regardless of whether the LLT terms are the same or different (per scenario b.1 and b.2); output NO if the PT terms match but dates don't overlap, or if PT terms differ entirely (per scenario b.3-b.4).

11. For each identified overlapping pair meeting the matching criteria, designate the record with the smaller Seq. No. as PROC 1 and the record with the larger Seq. No. as PROC 2.

12. Output one line per overlapping pair, including Country, Site, Subject Number, Form, Encoding Hierarchy, PROC Encoded Term, and the Seq/Verbatim/Start Date/Stop Date details for both PROC 1 and PROC 2.

13. Sort the final output by Country, Site, Subject Number, and PROC 1 Seq.

3. Spec Review

Readiness: Needs revision — The spec has good structural coverage of PT/LLT matching scenarios but contains a blocking ambiguity around the 'no parameter selected' matching behavior and an unaddressed cross-form (PROC vs APPRC) overlap edge case, plus several warnings around APPRC date substitution, partial date handling, and undocumented columns (Form, Encoding Hierarchy) that should be resolved before query generation."

BLOCKING (ambiguity): The Logic states 'User should select PT or LLT for comparison in parameter. If nothing is selected, output for overlapping identical LLT or PT.' It's unclear whether 'nothing selected' means the OR-logic of matching on either PT or LLT (i.e., match if either matches), or something else. The Report Scenarios section clarifies this differently by lumping 'no selection' together with 'PT selected' AND 'both selected' under the same behavior (section b), which uses PT-based matching (including cases where LLT differs). This contradicts a literal reading of 'overlapping identical LLT or PT' in the Logic, which could suggest an OR match on either term type. → Rewrite the no-selection behavior explicitly, e.g.: 'If no parameter is selected, treat this the same as PT selected (i.e., match on PT only, LLT need not match).' Remove the ambiguous 'or LLT' phrasing from the Logic sentence so it matches Report Scenario section b exactly.

WARNING (consistency): The Logic and Report Scenarios only discuss PT/LLT and start/stop dates as match criteria, and only mention 'Check within form only' for PROC and APPRC. But the Column Heading includes 'Encoding Hierarchy' and 'Form' without the Logic explaining what values populate these fields (e.g., is Form always 'PROC' or 'APPRC', and what does Encoding Hierarchy represent -- MedDRA hierarchy path, SOC, or just PT/LLT display?). → Add a line to the Logic describing exactly how 'Form' and 'Encoding Hierarchy' columns are derived (e.g., 'Form = PROC or APPRC depending on source', 'Encoding Hierarchy = SOC>HLGT>HLT>PT>LLT path from MedDRA').

BLOCKING (completeness): Report Scenarios say to 'check within form only' for PROC and APPRC, meaning overlaps are only compared within the same form. But the spec does not state what happens for overlaps across PROC and APPRC (e.g., a PROC record and an APPRC record for the same subject with the same PT/LLT and overlapping dates). This is an edge case that is implied (multiple forms exist) but not explicitly excluded or included in Report Scenarios. → Explicitly state whether cross-form (PROC vs APPRC) overlaps should ever be output, or add a scenario confirming 'PROC vs APPRC overlap = NO output' to remove ambiguity for the generator.

WARNING (completeness): The Logic says 'If either of the PROC has blank onset date or end date, consider them as ongoing,' implying ongoing records should still be evaluated for overlap. However, no Report Scenario explicitly demonstrates or confirms the expected YES/NO outcome for an ongoing record overlapping with another record (e.g., one record has no stop date and extends indefinitely). → Add a Report Scenario explicitly covering the ongoing-record case, e.g., 'PROC 1 has no stop date (ongoing) and overlaps with PROC 2's date range -> Output = YES', to make this edge case testable.

WARNING (ambiguity): The instruction 'For APPRC consider taking start date as stop date, since stop date is not available on form' is ambiguous: does it mean the start date value should populate both PROC Start Date AND PROC Stop Date columns for APPRC records, or is this only for internal overlap-calculation purposes without populating the actual Stop Date column (which might remain blank in output)? → Clarify: state explicitly whether the output columns 'PROC 1/2 Stop Date' should display the substituted start-date value for APPRC records, or remain blank while only being used internally for overlap comparison.

WARNING (ambiguity): 'Partial date also need to be considered while programming for identifying overlapping records' does not define how partial dates (e.g., year-only, year-month only, without UNK markers) should be handled/imputed, beyond the specific UNK conversion rules given afterward. It's unclear if 'partial' is synonymous with 'has UNK components' or includes other cases like a date with only year present. → Enumerate the specific partial date patterns expected (e.g., YYYY, YYYY-MM, YYYY-UNK-UNK) and the exact imputation rule for each, rather than relying on the general UNK conversion statement alone.

WARNING (testability): In section (b), scenario 2 ('same PT and different LLT after encoding have overlapping dates -> YES') and scenario 4 ('different Procedure terms (with different PT after encoding) -> NO') both involve 'different terms' at some level, but the spec doesn't clarify how a query generator/tester would distinguish 'different LLT' terms whose PT still matches vs. verifying PT truly differs. This is likely fine given the definitions, but the wording 'different Procedure terms (with different PT after encoding)' conflates term-level and code-level differences without an example, making scenario boundaries slightly hard to verify with sample data. → Add a brief concrete example

> (sample verbatim/PT/LLT trio) for each PT-selected scenario to make the distinctions unambiguous for test-case construction.

> **INFO (completeness):** The Logic states 'Deleted record and erroneous visits should not be considered' and Report Scenarios repeat 'Deleted PROC record -> NO', but there's no mention of how 'erroneous visits' are identified (e.g., a specific visit status/flag) nor a corresponding Report Scenario demonstrating exclusion of an erroneous-visit record. → Add a Report Scenario for erroneous visit exclusion, and specify the field/flag used to identify an 'erroneous visit' (e.g., Visit.Status = 'Erroneous').

4. Generated Code

CQL — Guardrails: passed

```
SELECT
  p1.SiteCountry AS `Country`,
  p1.SiteNumber AS `Site`,
  p1.SubjName AS `Subject Number`,
  p1.FormName AS `Form`,
  (CASE
     WHEN @LLTPT = 'LLT' THEN 'LLT'
     WHEN @LLTPT = 'PT' THEN 'PT'
     ELSE 'PT&LLT'
   END) AS `Encoding Hierarchy`,
  p1.EncodedTerm AS `PROC Encoded Term`,
  p1.Seq AS `PROC 1 Seq`,
  p1.Verbatim AS `PROC 1 Verbatim`,
  p1.StartDate AS `PROC 1 Start Date`,
  p1.StopDate AS `PROC 1 Stop Date`,
  p2.Seq AS `PROC 2 Seq`,
  p2.Verbatim AS `PROC 2 Verbatim`,
  p2.StartDate AS `PROC 2 Start Date`,
  p2.StopDate AS `PROC 2 Stop Date`
FROM
  (
    SELECT `@HDR`.`Subject`.`ID` AS SubjID,
      `@HDR`.`Site`.`Country` AS SiteCountry,
      `@HDR`.`Site`.`Number` AS SiteNumber,
      `@HDR`.`Subject`.`Name` AS SubjName,
      `@Form`.`Name` AS FormName,
      PROCPRAESEQIT AS Seq,
      PROCPRPROTRMIT AS Verbatim,
      CODEDTERM(PROCPRPROTRMIT, 'PT') AS PTTerm,
      CODEDTERM(PROCPRPROTRMIT, 'LLT') AS LLTTerm,
      (CASE
         WHEN @LLTPT = 'LLT' THEN CODEDTERM(PROCPRPROTRMIT, 'LLT')
         WHEN @LLTPT = 'PT' THEN CODEDTERM(PROCPRPROTRMIT, 'PT')
         ELSE CONCAT(CODEDTERM(PROCPRPROTRMIT, 'PT'), '/', CODEDTERM(PROCPRPROTRMIT, 'LLT'))
       END) AS EncodedTerm,
      RAWDATE(PROCPRPRSTDTCIT) AS StartDate,
      RAWDATE(PROCPRPRENDTCIT) AS StopDate
    FROM `EDC`.`PROCFM`
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (DELETEDFM = 'N' OR DELETEDFM IS NULL)
    UNION ALL
    SELECT `@HDR`.`Subject`.`ID` AS SubjID,
      `@HDR`.`Site`.`Country` AS SiteCountry,
      `@HDR`.`Site`.`Number` AS SiteNumber,
      `@HDR`.`Subject`.`Name` AS SubjName,
      `@Form`.`Name` AS FormName,
      ORAD2SEQNUMIT AS Seq,
      ORAD2PRTRTIT AS Verbatim,
      CODEDTERM(ORAD2PRTRTIT, 'PT') AS PTTerm,
      CODEDTERM(ORAD2PRTRTIT, 'LLT') AS LLTTerm,
      (CASE
         WHEN @LLTPT = 'LLT' THEN CODEDTERM(ORAD2PRTRTIT, 'LLT')
         WHEN @LLTPT = 'PT' THEN CODEDTERM(ORAD2PRTRTIT, 'PT')
         ELSE CONCAT(CODEDTERM(ORAD2PRTRTIT, 'PT'), '/', CODEDTERM(ORAD2PRTRTIT, 'LLT'))
       END) AS EncodedTerm,
      RAWDATE(ORAD2PRSTDTCIT) AS StartDate,
      RAWDATE(ORAD2PRSTDTCIT) AS StopDate
    FROM `EDC`.`ORAD2FM`
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (DELETEDFM = 'N' OR DELETEDFM IS NULL)
  ) p1
  JOIN
  (
    SELECT `@HDR`.`Subject`.`ID` AS SubjID,
      `@HDR`.`Site`.`Country` AS SiteCountry,
      `@HDR`.`Site`.`Number` AS SiteNumber,
      `@HDR`.`Subject`.`Name` AS SubjName,
      `@Form`.`Name` AS FormName,
      PROCPRAESEQIT AS Seq,
```

```
      PROCPRPROTRMIT AS Verbatim,
      CODEDTERM(PROCPRPROTRMIT, 'PT') AS PTTerm,
      CODEDTERM(PROCPRPROTRMIT, 'LLT') AS LLTTerm,
      RAWDATE(PROCPRPRSTDTCIT) AS StartDate,
      RAWDATE(PROCPRPRENDTCIT) AS StopDate
    FROM `EDC`.`PROCFM`
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (DELETEDFM = 'N' OR DELETEDFM IS NULL)
    UNION ALL
    SELECT `@HDR`.`Subject`.`ID` AS SubjID,
      `@HDR`.`Site`.`Country` AS SiteCountry,
      `@HDR`.`Site`.`Number` AS SiteNumber,
      `@HDR`.`Subject`.`Name` AS SubjName,
      `@Form`.`Name` AS FormName,
      ORAD2SEQNUMIT AS Seq,
      ORAD2PRTRTIT AS Verbatim,
      CODEDTERM(ORAD2PRTRTIT, 'PT') AS PTTerm,
      CODEDTERM(ORAD2PRTRTIT, 'LLT') AS LLTTerm,
      RAWDATE(ORAD2PRSTDTCIT) AS StartDate,
      RAWDATE(ORAD2PRSTDTCIT) AS StopDate
    FROM `EDC`.`ORAD2FM`
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (DELETEDFM = 'N' OR DELETEDFM IS NULL)
  ) p2
  ON p1.SubjID = p2.SubjID
  AND p1.FormName = p2.FormName
  AND p1.Seq < p2.Seq
WHERE
  (
    (@LLTPT = 'LLT' AND p1.LLTTerm = p2.LLTTerm)
    OR
    (@LLTPT <> 'LLT' AND p1.PTTerm = p2.PTTerm)
  )
  AND p1.StartDate <= IFNULL(p2.StopDate, p1.StartDate)
  AND p2.StartDate <= IFNULL(p1.StopDate, p2.StartDate)
ORDER BY `Country`, `Site`, `Subject Number`, `PROC 1 Seq`
```

Following the near-identical prior spec pattern, I built a UNION ALL derived table combining the PROCFM ("PROC") form and ORAD2FM (treated as "APPRC", since ORAD2FM is the only other procedure-like form with start date, PT-eligible verbatim term item ORAD2PRTRTIT, and no stop date item -- start date is reused as stop date per the "For APPRC consider taking start date as stop date" rule); this union is self-joined (aliased p1/p2) on Subject ID + FormName (to enforce "check within form only") with p1.Seq < p2.Seq to avoid duplicate/reversed pairs. The @LLTPT parameter (assumed name, user-selectable PT/LLT per spec) drives both the encoding-hierarchy label/term match logic (LLT-only match when 'LLT' selected, otherwise PT match per spec's scenario rules) and the displayed Encoded Term (LLT, PT, or PT/LLT concatenation when neither/both selected). Overlap is computed via IFNULL-based date comparison treating missing start/end as ongoing; deleted records are excluded via DELETEDFM tolerant of NULL, and RAWDATE() is used to normalize partial dates. Please confirm via DESCRIBE FORM that ORAD2FM is indeed the intended "APPRC" form and that ORAD2PRTRTIT/ORAD2SEQNUMIT are the correct verbatim/seq items, since APPRC wasn't explicitly present in the provided metadata list.